# Extended E2 (E2X): results

Built by `scripts/e2x/make_report.py` from the files listed at the end. Preregistration: tag `e2x-prereg` (`experiments/E2X.md` §5). DESCRIPTIVE parts are labelled; calibration-clip numbers are not results.

> **ANALYSIS PAPER (no method claim): E3 is NOT CALIBRATED for MapAnything, so the photo witness cannot be made safe on correct hints under the fixed rule. E3 is reported as a negative result.**

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## Decisions (rules fixed in advance)

### E2X continue / stop rule (E2X.md §5; primary calibration)

| Signal |
|---|
| (i-a) real whole-room pose hints HURT for ≥ 1 model (X-B, dev_pose, rotation or shape) |
| (i-b) E3 removes more damage than the best of R0/R2/R3 at ≥ its benefit kept (MapAnything, cal clips; R1 'drop all poses' is not in the |
| (ii) E3 calibrated for MapAnything |

> **ANALYSIS PAPER (no method claim): E3 is NOT CALIBRATED for MapAnything, so the photo witness cannot be made safe on correct hints under the fixed rule. E3 is reported as a negative result.**

### E3 method evaluation (E3.md; thresholds from the accurate-ground-truth calibration)

> **E3 method claim holds: False (SAFE: True). Both readings are reported: the primary rule above and this evaluation.**

Inside-the-network direction: Shubh's G1 oracle dial gave no reading in session A (its own γ = 1 check failed). Session A2 measured the same dial (M1, section 6).

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## 1. The session

```
NVIDIA A100-SXM4-80GB, 580.178.04, 81920 MiB, 0 MiB
cpus 12
Mem:             167           5         112           0          50         161
/dev/root       991G   66G  926G   7% /
torch wheels cu130
our GPU budget 32 GB
```

First and last queue lines:

```
2026-11-07 18:38:51 start e2x session A, budget 32 GB of 80 GB, free now 15 GB, tree
ab5bc73d5a844da74539d390883b6273ad6b3482, scripts 1c0835f8c12bac528b08c2d16a6de124d1f0fb6c
2026-11-08 08:48:59 PART_PX_DONE
```

AI depth check against E2-E's record: `AI depth on the E2-E clips: 0/49 bit-identical to E2-E; same model revision: True`

### Runs per job

| run | rows | ok | failed rows | failures.jsonl lines | out-of-memory resets |
|---|---|---|---|---|---|
| e2x_e2e_px_dev | 168 | 168 | 0 | 0 | 0 |
| e2x_e2e_px_eval | 861 | 861 | 0 | 0 | 0 |
| e2x_e3cal_real_ma_cal | 2050 | 2050 | 0 | 0 | 0 |
| e2x_e3cal_real_ma_roomdev | 287 | 287 | 0 | 0 | 0 |
| e2x_e3cal_real_px_cal | 2050 | 2050 | 0 | 0 | 0 |
| e2x_e3cal_synth_ma_cal | 16000 | 16000 | 0 | 0 | 27 |
| e2x_e3cal_synth_ma_eth3d | 960 | 960 | 0 | 0 | 0 |
| e2x_e3cal_synth_ma_tum | 209 | 209 | 0 | 0 | 0 |
| e2x_e3cal_synth_ov_eth3d | 960 | 960 | 0 | 0 | 0 |
| e2x_e3cal_synth_px_cal | 16000 | 16000 | 0 | 0 | 0 |
| e2x_e3cal_synth_px_eth3d | 960 | 960 | 0 | 0 | 0 |
| e2x_e3cal_synth_px_tum | 209 | 209 | 0 | 0 | 0 |
| e2x_pi3x_checks | 129 | 129 | 0 | 0 | 0 |
| e2x_pi3x_eth3d_dev | 1386 | 1386 | 0 | 0 | 0 |
| e2x_pi3x_eth3d_eval | 11088 | 11088 | 0 | 0 | 0 |
| e2x_room_ma_dev | 147 | 147 | 0 | 0 | 0 |
| e2x_room_ma_eval | 756 | 756 | 0 | 0 | 0 |
| e2x_room_ov_dev | 133 | 133 | 0 | 0 | 0 |
| e2x_room_ov_eval | 684 | 684 | 0 | 0 | 0 |
| e2x_room_px_dev | 147 | 147 | 0 | 0 | 0 |
| e2x_room_px_eval | 756 | 756 | 0 | 0 | 0 |

### Dev gates and job markers

```
2026-11-07 19:06:08 GATE_MA had errors: see /home/dawar/e2x_logs/gate.txt (queue continues)
GATE e2e results/e2x_room_ov_dev: PASS
GATE ov_e2d results/e2x_pi3x_eth3d_dev: PASS
GATE e2e results/e2x_room_ma_dev: PASS
GATE e2e results/e2x_e2e_px_dev: PASS
GATE e2e results/e2x_room_px_dev: PASS
2026-11-08 01:27:07   !! 27 runs reset after CUDA out-of-memory; all processes ok: 1 (logs:
results/e2x_e3cal_synth_ma_cal/_shards/*/log.txt). Retry in 120 s
2026-11-08 01:40:35   Pi3X peak measured: 9 GB + headroom -> 10 GB per process
2026-11-08 02:38:20 ROOM_OV_EVAL_DONE
2026-11-08 03:22:04 ROOM_MA_EVAL_DONE
```

```
2026-11-08 03:26:37 PX_ETH3D_EVAL_DONE
2026-11-08 03:59:25 PX_E2E_EVAL_DONE
2026-11-08 04:31:03 ROOM_PX_EVAL_DONE
2026-11-08 08:48:59 PART_PX_DONE
```

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## 2. X-A: Pi3X, the third model (ETH3D)

Preregistered D-H test (E2-D verbatim, Holm within the model): **5** of 56 tests harmful, families: depth_one_scale, drift, outlier_frac, outlier_rot.

| condition | metric | x_mean | floor_mean | delta | ci_lo | ci_hi | n_worse | p_holm |
|---|---|---|---|---|---|---|---|---|
| depth_one_scale_factor=1.5 | shape | 0.071 | 0.021 | 0.051 | 0.041 | 0.060 | 10 | 0.0002 |
| outlier_rot_rot_deg=90 | rotation | 8.373 | 0.491 | 7.882 | 6.002 | 9.824 | 10 | 0.0018 |
| outlier_rot_rot_deg=30 | rotation | 1.337 | 0.491 | 0.846 | 0.579 | 1.103 | 10 | 0.0114 |
| drift_level=3 | rotation | 1.910 | 0.491 | 1.419 | 0.976 | 1.883 | 10 | 0.0133 |
| outlier_frac_frac=0.4 | rotation | 2.007 | 0.491 | 1.516 | 1.052 | 2.058 | 10 | 0.0192 |

### Behaviour (P1–P5, P10, P11)

| test | question | mean | ci_lo | ci_hi | n_scenes | p_holm | finding |
|---|---|---|---|---|---|---|---|
| P1_rot_override | C1 | 0.082 | 0.027 | 0.145 | 10 | 0.1559 | False |
| P2_trans_override | C1 | 0.446 | 0.347 | 0.539 | 10 | 0.0001 | False |
| P3_conf_flags_bad_view | C2 | -0.015 | -0.106 | 0.081 | 10 | 0.7823 | False |
| P4_bad_view_own_shape | C3 | 0.000 | 0.000 | 0.001 | 10 | 0.1803 | False |
| P5_good_views_shape | C3 | 0.000 | 0.000 | 0.000 | 10 | 0.1803 | False |
| P10_shared_vs_indep_conf_5deg | C6 | 0.004 | 0.002 | 0.007 | 10 | 0.1065 | False |
| P11_drift_vs_jitter_conf_level2 | C6 | 0.001 | 0.000 | 0.002 | 10 | 0.1803 | False |

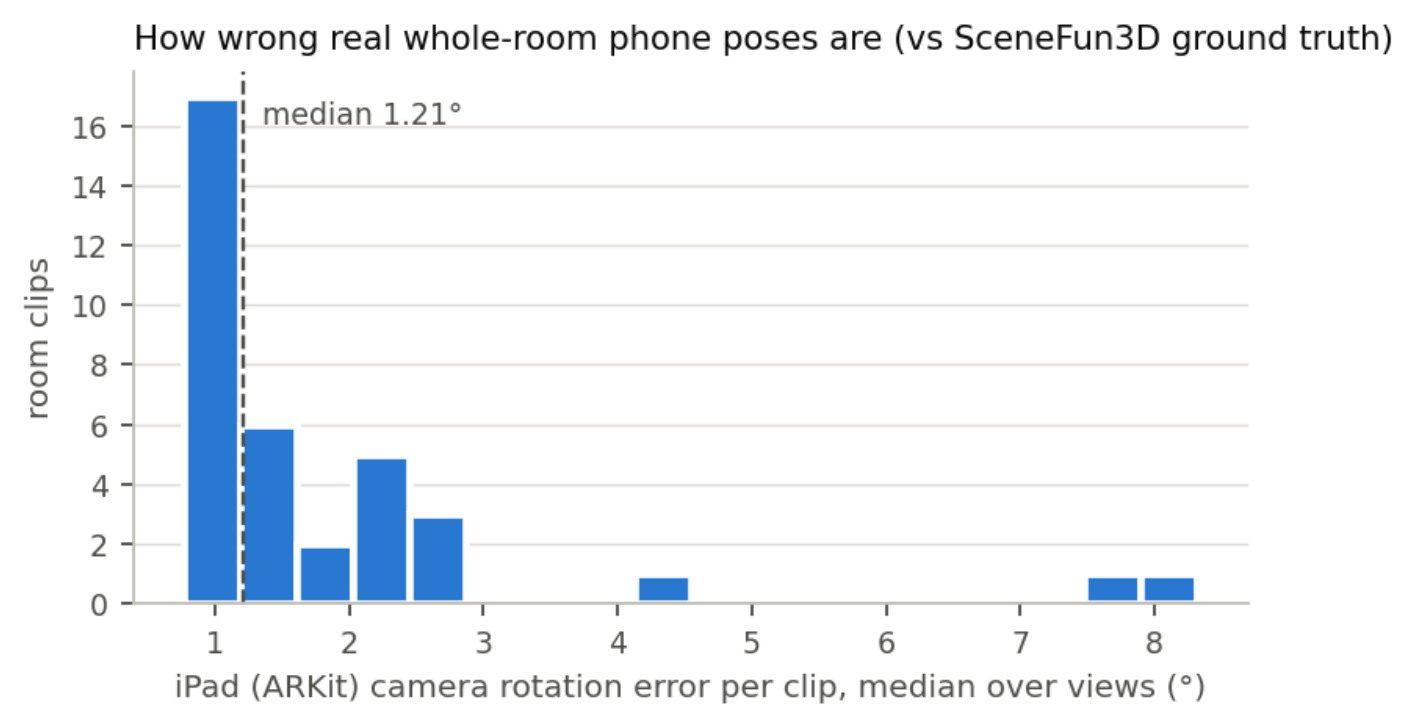
### Per family: MapAnything vs Pi3X

| family | class |
|---|---|
| anchor_error | neither |
| depth_one_scale | same |
| depth_scale | MapAnything only |
| drift | same |
| noise_rot | neither |
| outlier_frac | same |
| outlier_rot | same |
| scale_all | MapAnything only |
| shared_heading_bias | neither |

### Behaviour side by side

| test | ma_mean | ma_finding | ov_mean | ov_finding | class |
|---|---|---|---|---|---|
| P1_rot_override | 0.028 | False | 0.082 | False | neither |
| P2_trans_override | 0.112 | False | 0.446 | False | neither |
| P3_conf_flags_bad_view | 0.431 | True | -0.015 | False | MapAnything only |
| P4_bad_view_own_shape | 0.006 | False | 0.000 | False | neither |
| P5_good_views_shape | 0.000 | False | 0.000 | False | neither |
| P10_shared_vs_indep_conf_5deg | 0.252 | True | 0.004 | False | MapAnything only |
| P11_drift_vs_jitter_conf_level2 | 0.280 | True | 0.001 | False | MapAnything only |

## 3. X-B: real hints over whole rooms




36 room clips: median rotation error 1.21° (clip medians), share of views ≥ 5° 0.06, depth scale ratio LiDAR/GT median 0.995 (`results/e2x_agg/xb_room_real_errors.csv`).

**MapAnything: each real hint set vs images only (room eval clips)**

| condition | metric | delta | ci_lo | ci_hi | n_scenes | p_holm | verdict |
|---|---|---|---|---|---|---|---|
| dev_K | shape | -0.011 | -0.022 | 0.000 | 15 | 0.6777 | no clear effect |
| dev_K | rotation | 0.213 | -0.794 | 1.490 | 15 | 1.0000 | no clear effect |
| dev_pose | shape | -0.162 | -0.292 | -0.072 | 15 | 0.1536 | no clear effect |
| dev_pose | rotation | -15.210 | -24.785 | -6.760 | 15 | 0.0953 | no clear effect |
| dev_pose | real_size | -0.449 | -0.619 | -0.323 | 15 | 0.0011 | HELPS |
| dev_depth | shape | -0.072 | -0.165 | -0.014 | 15 | 0.7267 | no clear effect |
| dev_depth | rotation | -4.018 | -9.456 | -0.021 | 15 | 0.7267 | no clear effect |
| dev_depth | real_size | -0.369 | -0.506 | -0.266 | 15 | 0.0011 | HELPS |
| dev_all | shape | -0.186 | -0.322 | -0.089 | 15 | 0.1166 | no clear effect |
| dev_all | rotation | -15.220 | -24.784 | -6.776 | 15 | 0.0953 | no clear effect |
| dev_all | real_size | -0.482 | -0.658 | -0.349 | 15 | 0.0009 | HELPS |
| ai_depth | shape | 0.053 | -0.046 | 0.132 | 15 | 1.0000 | no clear effect |
| ai_depth | rotation | -4.236 | -9.116 | -0.216 | 15 | 0.6924 | no clear effect |
| ai_depth | real_size | -0.039 | -0.178 | 0.095 | 15 | 1.0000 | no clear effect |
| dev_pose_ai_depth | shape | -0.014 | -0.138 | 0.078 | 15 | 1.0000 | no clear effect |
| dev_pose_ai_depth | rotation | -15.221 | -24.780 | -6.776 | 15 | 0.0953 | no clear effect |
| dev_pose_ai_depth | real_size | -0.231 | -0.399 | -0.112 | 15 | 0.1166 | no clear effect |

Remedies on the real hints (Δ vs no remedy):

| condition | remedy | metric | delta_vs_no_remedy | ci_lo | ci_hi | share_views_dropped |
|---|---|---|---|---|---|---|
| dev_pose | filter_rgbcheck | shape | 0.137 | 0.045 | 0.272 | 0.321 |
| dev_pose | filter_rgbcheck | rotation | 11.915 | 4.156 | 20.945 | 0.321 |
| dev_pose | drop_lowconf | shape | 0.006 | 0.001 | 0.013 | 0.037 |
| dev_pose | drop_lowconf | rotation | 0.866 | 0.030 | 2.454 | 0.037 |
| dev_all | filter_rgbcheck | shape | 0.096 | 0.042 | 0.160 | 0.321 |
| dev_all | filter_rgbcheck | rotation | 9.455 | 3.760 | 16.122 | 0.321 |
| dev_all | drop_lowconf | shape | 0.007 | 0.002 | 0.012 | 0.074 |
| dev_all | drop_lowconf | rotation | 1.850 | 0.346 | 3.770 | 0.074 |
| dev_pose_ai_depth | filter_rgbcheck | shape | 0.053 | 0.017 | 0.096 | 0.321 |
| dev_pose_ai_depth | filter_rgbcheck | rotation | 7.915 | 3.177 | 13.494 | 0.321 |
| dev_pose_ai_depth | drop_lowconf | shape | -0.007 | -0.011 | -0.003 | 0.086 |
| dev_pose_ai_depth | drop_lowconf | rotation | 1.760 | 0.410 | 3.590 | 0.086 |

### OmniVGGT: each real hint set vs images only (room eval clips)

| condition | metric | delta | ci_lo | ci_hi | n_scenes | p_holm | verdict |
|---|---|---|---|---|---|---|---|
| dev_pose | shape | -0.130 | -0.255 | -0.039 | 15 | 0.2606 | no clear effect |
| dev_pose | rotation | -11.757 | -20.282 | -4.504 | 15 | 0.1246 | no clear effect |
| dev_depth | shape | 0.028 | -0.004 | 0.066 | 15 | 0.4450 | no clear effect |
| dev_depth | rotation | 0.695 | -2.052 | 2.897 | 15 | 1.0000 | no clear effect |
| dev_all | shape | -0.122 | -0.243 | -0.037 | 15 | 0.2606 | no clear effect |
| dev_all | rotation | -11.749 | -20.287 | -4.521 | 15 | 0.1246 | no clear effect |
| ai_depth | shape | 0.128 | 0.089 | 0.178 | 15 | 0.0010 | HURTS |
| ai_depth | rotation | 2.935 | 0.155 | 5.965 | 15 | 0.2923 | no clear effect |
| dev_pose_ai_depth | shape | -0.038 | -0.157 | 0.046 | 15 | 1.0000 | no clear effect |
| dev_pose_ai_depth | rotation | -11.502 | -20.048 | -4.264 | 15 | 0.1246 | no clear effect |

Remedies on the real hints (Δ vs no remedy):

| condition | remedy | metric | delta_vs_no_remedy | ci_lo | ci_hi | share_views_dropped |
|---|---|---|---|---|---|---|
| dev_pose | filter_rgbcheck | shape | 0.097 | 0.018 | 0.196 | 0.225 |
| dev_pose | filter_rgbcheck | rotation | 9.243 | 3.082 | 16.669 | 0.225 |
| dev_pose | drop_lowconf | shape | 0.000 | -0.000 | 0.000 | 0.009 |
| dev_pose | drop_lowconf | rotation | 0.017 | 0.000 | 0.036 | 0.009 |
| dev_all | filter_rgbcheck | shape | 0.086 | 0.021 | 0.171 | 0.225 |
| dev_all | filter_rgbcheck | rotation | 10.443 | 4.125 | 18.110 | 0.225 |
| dev_all | drop_lowconf | shape | 0.000 | 0.000 | 0.000 | 0.000 |
| dev_all | drop_lowconf | rotation | 0.000 | 0.000 | 0.000 | 0.000 |
| dev_pose_ai_depth | filter_rgbcheck | shape | 0.087 | 0.023 | 0.170 | 0.225 |
| dev_pose_ai_depth | filter_rgbcheck | rotation | 10.996 | 4.868 | 18.299 | 0.225 |
| dev_pose_ai_depth | drop_lowconf | shape | -0.000 | -0.001 | 0.000 | 0.003 |
| dev_pose_ai_depth | drop_lowconf | rotation | 0.016 | 0.000 | 0.049 | 0.003 |

### Pi3X: each real hint set vs images only (room eval clips)

| condition | metric | delta | ci_lo | ci_hi | n_scenes | p_holm | verdict |
|---|---|---|---|---|---|---|---|
| dev_K | shape | 0.014 | -0.008 | 0.040 | 15 | 0.8173 | no clear effect |
| dev_K | rotation | 0.504 | -0.106 | 1.378 | 15 | 0.8173 | no clear effect |
| dev_pose | shape | -0.162 | -0.289 | -0.056 | 15 | 0.1554 | no clear effect |
| dev_pose | rotation | -6.375 | -10.917 | -2.559 | 15 | 0.1260 | no clear effect |
| dev_depth | shape | 0.073 | -0.022 | 0.187 | 15 | 0.8173 | no clear effect |
| dev_depth | rotation | 2.580 | -0.325 | 6.029 | 15 | 0.7634 | no clear effect |
| dev_all | shape | -0.165 | -0.293 | -0.057 | 15 | 0.1554 | no clear effect |
| dev_all | rotation | -6.515 | -11.023 | -2.729 | 15 | 0.1227 | no clear effect |
| ai_depth | shape | 0.200 | 0.109 | 0.307 | 15 | 0.0230 | HURTS |
| ai_depth | rotation | 5.547 | 2.184 | 9.386 | 15 | 0.1260 | no clear effect |
| dev_pose_ai_depth | shape | -0.000 | -0.123 | 0.103 | 15 | 0.9989 | no clear effect |
| dev_pose_ai_depth | rotation | -5.796 | -10.365 | -1.981 | 15 | 0.1554 | no clear effect |

Remedies on the real hints (Δ vs no remedy):

| condition | remedy | metric | delta_vs_no_remedy | ci_lo | ci_hi | share_views_dropped |
|---|---|---|---|---|---|---|
| dev_pose | filter_rgbcheck | shape | 0.183 | 0.052 | 0.341 | 0.290 |
| dev_pose | filter_rgbcheck | rotation | 6.783 | 2.467 | 11.719 | 0.290 |
| dev_pose | drop_lowconf | shape | 0.000 | 0.000 | 0.000 | 0.000 |
| dev_pose | drop_lowconf | rotation | 0.000 | 0.000 | 0.000 | 0.000 |
| dev_all | filter_rgbcheck | shape | 0.225 | 0.034 | 0.457 | 0.290 |
| dev_all | filter_rgbcheck | rotation | 8.018 | 2.338 | 14.897 | 0.290 |
| dev_all | drop_lowconf | shape | -0.000 | -0.000 | 0.000 | 0.003 |
| dev_all | drop_lowconf | rotation | -0.000 | -0.001 | 0.000 | 0.003 |
| dev_pose_ai_depth | filter_rgbcheck | shape | 0.205 | 0.037 | 0.413 | 0.290 |
| dev_pose_ai_depth | filter_rgbcheck | rotation | 9.704 | 3.682 | 16.628 | 0.290 |
| dev_pose_ai_depth | drop_lowconf | shape | 0.000 | 0.000 | 0.000 | 0.000 |
| dev_pose_ai_depth | drop_lowconf | rotation | 0.000 | 0.000 | 0.000 | 0.000 |

### Pi3X on the E2-E short clips

| condition | metric | delta | ci_lo | ci_hi | n_scenes | p_holm | verdict |
|---|---|---|---|---|---|---|---|
| dev_K | shape | -0.010 | -0.016 | -0.006 | 15 | 0.0126 | HELPS |
| dev_K | rotation | -0.231 | -0.305 | -0.163 | 15 | 0.0002 | clear but below the practical threshold |
| dev_pose | shape | -0.012 | -0.020 | -0.006 | 15 | 0.0130 | HELPS |
| dev_pose | rotation | -0.297 | -0.386 | -0.216 | 15 | 0.0001 | HELPS |
| dev_depth | shape | -0.004 | -0.013 | 0.006 | 15 | 0.4037 | no clear effect |
| dev_depth | rotation | -0.159 | -0.277 | -0.019 | 15 | 0.0689 | no clear effect |
| dev_all | shape | -0.010 | -0.018 | -0.004 | 15 | 0.0463 | HELPS |
| dev_all | rotation | -0.300 | -0.391 | -0.218 | 15 | 0.0001 | HELPS |
| ai_depth | shape | 0.115 | 0.086 | 0.146 | 15 | 0.0000 | HURTS |
| ai_depth | rotation | 1.136 | 0.935 | 1.343 | 15 | 0.0000 | HURTS |
| dev_pose_ai_depth | shape | 0.106 | 0.085 | 0.128 | 15 | 0.0000 | HURTS |
| dev_pose_ai_depth | rotation | 0.260 | 0.123 | 0.393 | 15 | 0.0130 | HURTS |

## 4. E3 calibration (calibration and dev clips only; nothing here is a result)

Rule fixed before the session (E2X.md §5): the smallest `e3_z` whose false-action rate on ground-truth hints is ≤ 5% for any group action and ≤ 2% of views for single-view drops; the same for the depth cut-offs; otherwise NOT CALIBRATED, never loosened.

### MapAnything

Chosen: `{"e3_z": "NOT CALIBRATED", "e3_tau_depth_shared": 0.2, "e3_tau_depth_view": 0.25, "n_gt_pose_decisions": 269, "n_gt_depth_decisions": 110}`

| part | setting | n | group_action_rate | view_drop_rate | meets_rule |
|---|---|---|---|---|---|
| pose | e3_z=2.0 | 269 | 0.599 | 0.118 | False |
| pose | e3_z=2.5 | 269 | 0.461 | 0.061 | False |
| pose | e3_z=3.0 | 269 | 0.398 | 0.044 | False |
| pose | e3_z=4.0 | 269 | 0.249 | 0.038 | False |
| pose | e3_z=5.0 | 269 | 0.138 | 0.033 | False |
| depth_shared | e3_tau_depth_shared=0.05 | 110 | 0.518 | – | False |
| depth_shared | e3_tau_depth_shared=0.1 | 110 | 0.255 | – | False |
| depth_shared | e3_tau_depth_shared=0.15 | 110 | 0.100 | – | False |
| depth_shared | e3_tau_depth_shared=0.2 | 110 | 0.036 | – | True |
| depth_view | e3_tau_depth_view=0.1 | 110 | – | 0.089 | False |
| depth_view | e3_tau_depth_view=0.15 | 110 | – | 0.046 | False |
| depth_view | e3_tau_depth_view=0.25 | 110 | – | 0.019 | True |

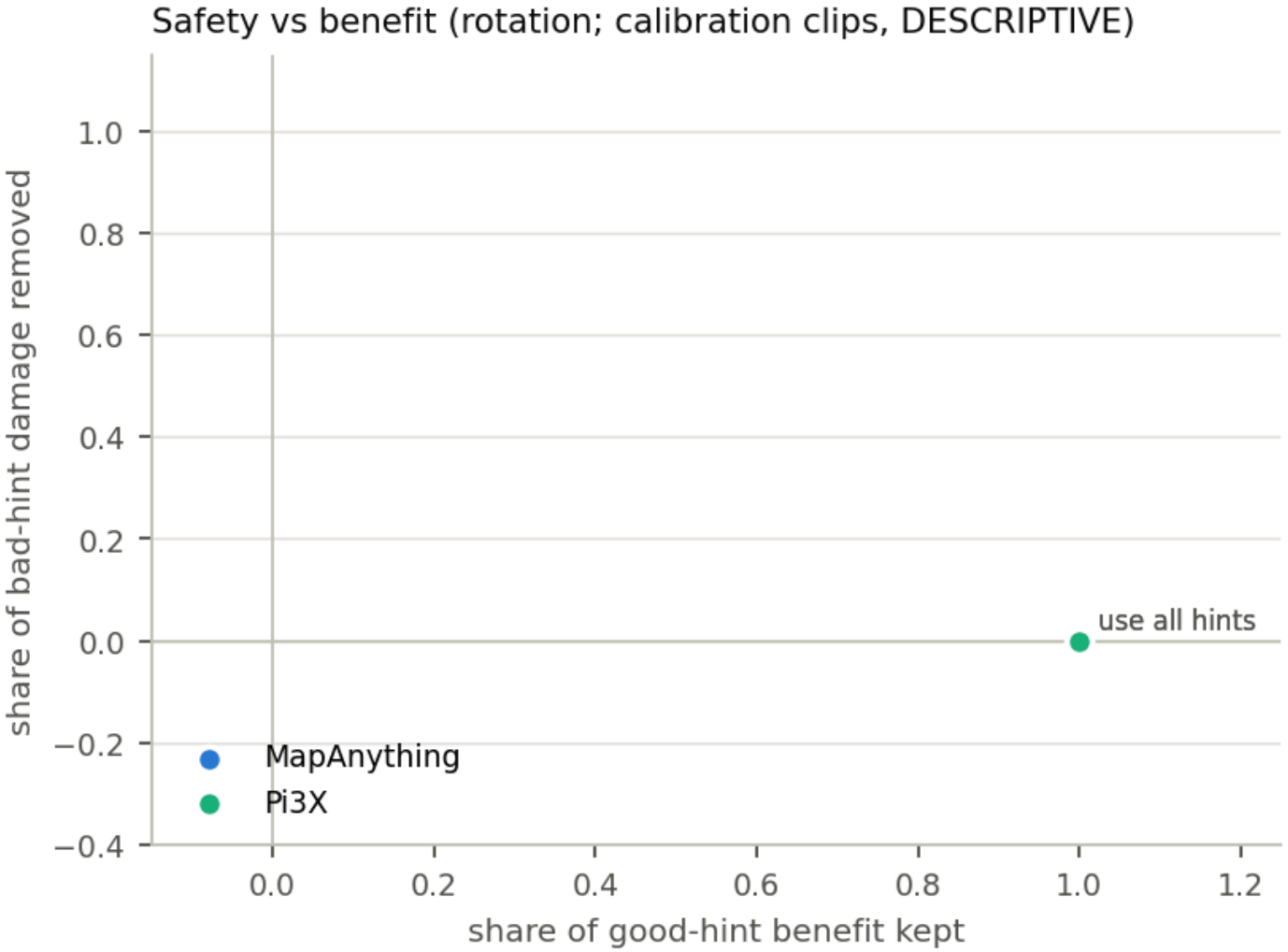
### Pi3X

Chosen: {"e3_z": "NOT CALIBRATED", "e3_tau_depth_shared": 0.15, "e3_tau_depth_view": 0.15, "n_gt_pose_decisions": 206, "n_gt_depth_decisions": 103}

| part | setting | n | group_action_rate | view_drop_rate | meets_rule |
|---|---|---|---|---|---|
| pose | e3_z=2.0 | 206 | 0.786 | 0.157 | False |
| pose | e3_z=2.5 | 206 | 0.757 | 0.119 | False |
| pose | e3_z=3.0 | 206 | 0.680 | 0.104 | False |
| pose | e3_z=4.0 | 206 | 0.485 | 0.091 | False |
| pose | e3_z=5.0 | 206 | 0.388 | 0.076 | False |
| depth_shared | e3_tau_depth_shared=0.05 | 103 | 0.272 | – | False |
| depth_shared | e3_tau_depth_shared=0.1 | 103 | 0.087 | – | False |
| depth_shared | e3_tau_depth_shared=0.15 | 103 | 0.039 | – | True |
| depth_shared | e3_tau_depth_shared=0.2 | 103 | 0.039 | – | True |
| depth_view | e3_tau_depth_view=0.1 | 103 | – | 0.036 | False |
| depth_view | e3_tau_depth_view=0.15 | 103 | – | 0.017 | True |
| depth_view | e3_tau_depth_view=0.25 | 103 | – | 0.007 | True |

### OmniVGGT

*Not available: results/e2x_e3cal_agg/omnivggt/chosen.json (input file missing; nothing is estimated).*


Safety vs benefit (rotation; calibration clips, DESCRIPTIVE)
share of bad-hint damage removed
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
use all hints
MapAnything
Pi3X
0.0
0.2
0.4
0.6
0.8
1.0
1.2
share of good-hint benefit kept

| model | metric | remedy | benefit_kept | damage_removed | n_bad_conditions |
|---|---|---|---|---|---|
| mapanything | rre_deg | drop_lowconf | 0.938 | -1.471 | 21 |
| mapanything | rre_deg | filter_rgbcheck | 0.321 | -41.025 | 21 |
| mapanything | rre_deg | none | 1.000 | 0.000 | 21 |
| mapanything | rre_deg | witness | 0.550 | -23.558 | 21 |
| mapanything | rre_deg | witness_drop | 0.358 | -33.489 | 21 |
| mapanything | rre_deg | R0 images only | 0.000 | -49.565 | – |
| mapanything | pts_relerr_scale | drop_lowconf | 0.964 | -4.271 | 21 |
| mapanything | pts_relerr_scale | filter_rgbcheck | 0.371 | -70.622 | 21 |
| mapanything | pts_relerr_scale | none | 1.000 | 0.000 | 21 |
| mapanything | pts_relerr_scale | witness | 0.623 | -38.318 | 21 |
| mapanything | pts_relerr_scale | witness_drop | 0.353 | -63.249 | 21 |
| mapanything | pts_relerr_scale | R0 images only | 0.000 | -99.402 | – |
| pi3x | rre_deg | drop_lowconf | 0.987 | -1.279 | 21 |
| pi3x | rre_deg | filter_rgbcheck | 0.274 | -37.356 | 21 |
| pi3x | rre_deg | none | 1.000 | 0.000 | 21 |
| pi3x | rre_deg | witness | 0.237 | -34.644 | 21 |
| pi3x | rre_deg | witness_drop | 0.048 | -39.734 | 21 |
| pi3x | rre_deg | R0 images only | 0.000 | -41.074 | – |
| pi3x | pts_relerr_scale | drop_lowconf | 0.984 | -5.715 | 21 |
| pi3x | pts_relerr_scale | filter_rgbcheck | 0.399 | -134.425 | 21 |
| pi3x | pts_relerr_scale | none | 1.000 | 0.000 | 21 |
| pi3x | pts_relerr_scale | witness | 0.456 | -135.224 | 21 |
| pi3x | pts_relerr_scale | witness_drop | 0.211 | -165.511 | 21 |
| pi3x | pts_relerr_scale | R0 images only | 0.000 | -177.009 | – |

## 5b. E3 calibration on accurate ground truth (E3-cal amendment 1, SECONDARY)

Written before any TUM run (tag `e3cal-amend1`): the same rule on ETH3D dev (laser scan) and TUM RGB-D fr3 (motion capture) clips; TUM clips failing the E0 ground-truth check were excluded first (`results/e2x_tum_clip_check/excluded.csv`). The primary verdict (section 4) stands.

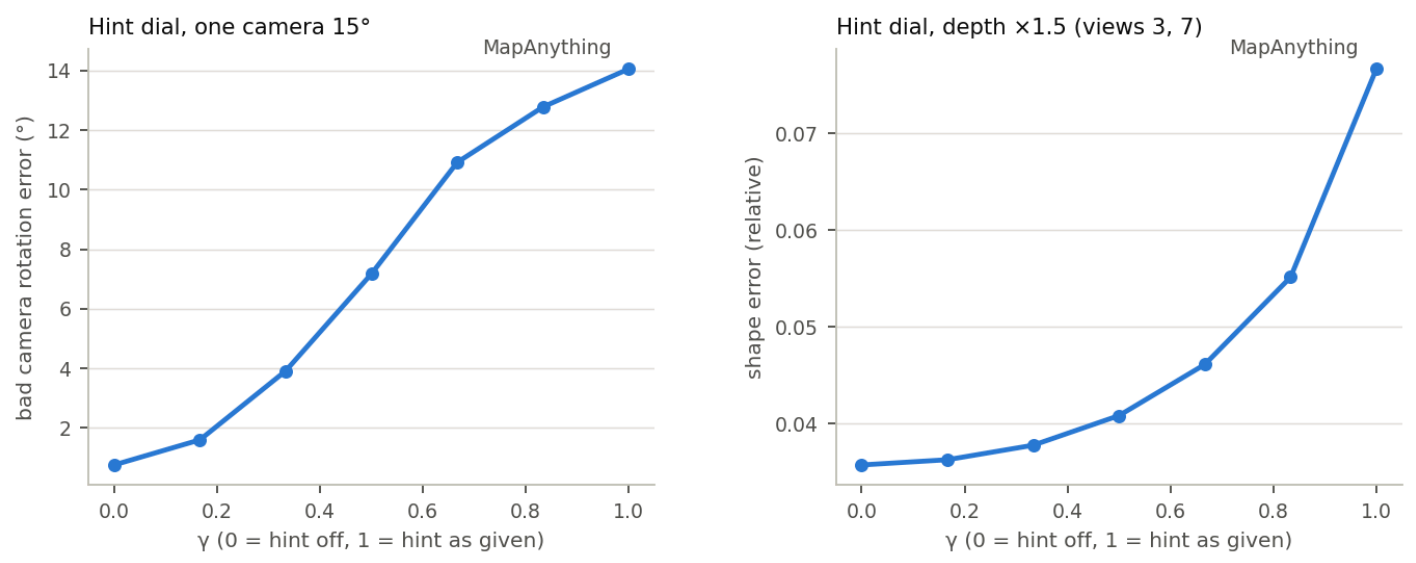
| model | calibration | e3_z | tau_depth_shared | tau_depth_view | n_gt_pose_decisions | n_gt_depth_de |
|---|---|---|---|---|---|---|
| MapAnything | primary | NOT CALIBRATED | 0.2 | 0.25 | 269 | |
| MapAnything | secondary | 5.0 | 0.1 | 0.1 | 44 | |
| Pi3X | primary | NOT CALIBRATED | 0.15 | 0.15 | 206 | |
| Pi3X | secondary | 4.0 | 0.05 | 0.1 | 44 | |

▶ MapAnything: secondary false-action rates

▶ Pi3X: secondary false-action rates

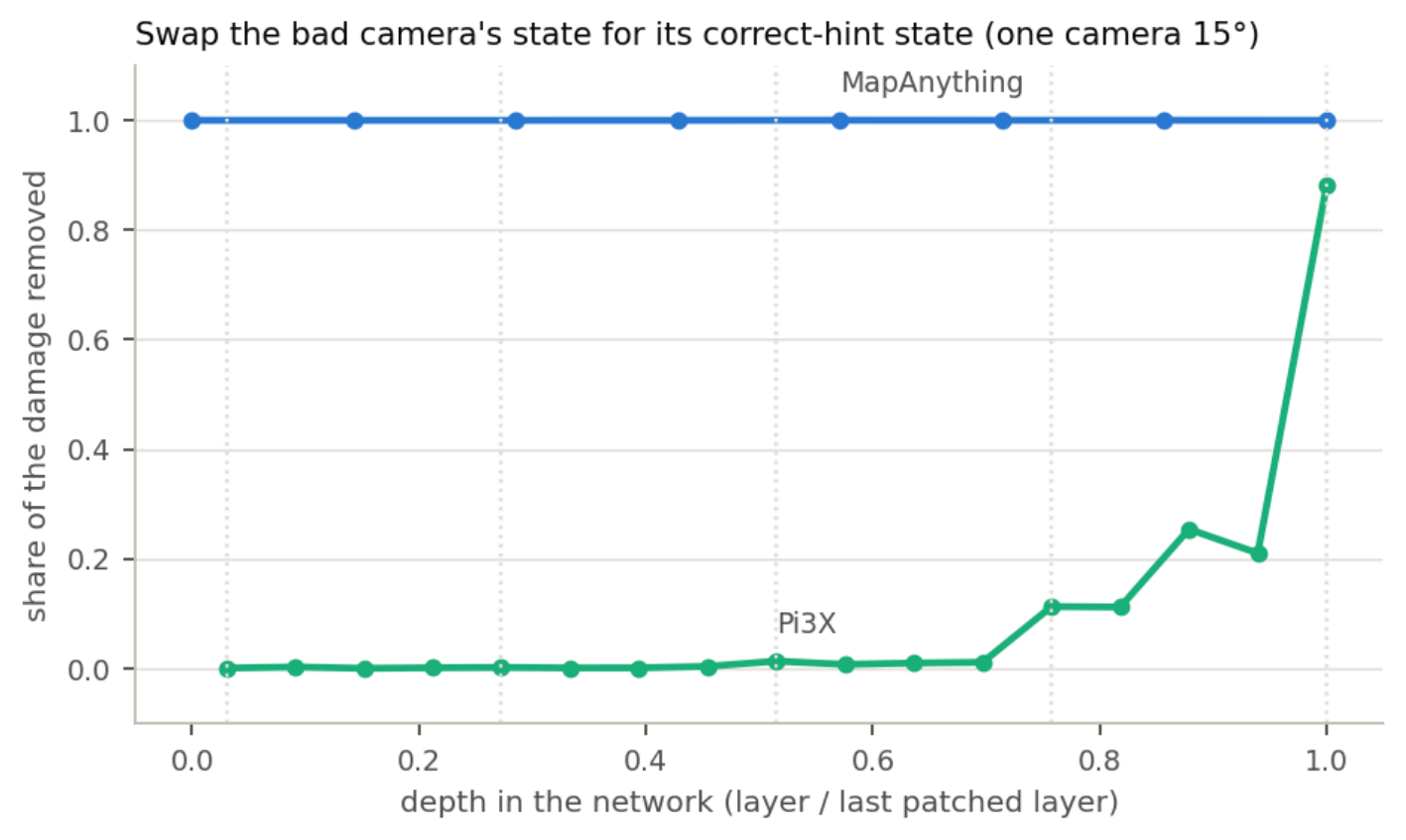
## 6. Inside the network (session A2, DESCRIPTIVE; ETH3D eval, 24 clips, 10 scenes)

Frozen at `e2x-a2-prereg` (amendments 1–2: plumbing fixes before any eval run). Pi3X injects the camera hint again after blocks 1, 9, 17, 25, 33, so a swapped state can be overwritten later (stated before the run); the dotted lines mark those blocks.




### M1 hint dial

| model | case | r2_mean | r2_ci_lo | r2_ci_hi | n_scenes_linear | switch_index_mean | n_scenes_switch |
|---|---|---|---|---|---|---|---|
| mapanything | depth_x1.5 | 0.678 | 0.534 | 0.785 | 0 | -0.013 | 0 |
| mapanything | drift_level2 | 0.595 | 0.397 | 0.776 | 1 | 0.143 | 1 |
| mapanything | one_cam_15 | 0.944 | 0.914 | 0.971 | 7 | 0.063 | 0 |
| mapanything | shared_heading_5 | 0.634 | 0.461 | 0.785 | 0 | 0.154 | 1 |




**M3 point of no return (first layer whose swap removes < 50% of the damage)**

| model | case | source | point_of_no_return_layer |
|---|---|---|---|
| mapanything | depth_x1.5 | clean | 8.0 |
| mapanything | depth_x1.5 | drop | 8.0 |
| mapanything | drift_level2 | clean | none |
| mapanything | drift_level2 | drop | 8.0 |
| mapanything | one_cam_15 | clean | none |
| mapanything | one_cam_15 | drop | none |
| mapanything | shared_heading_5 | clean | none |
| mapanything | shared_heading_5 | drop | 8.0 |
| pi3x | depth_x1.5 | clean | 1.0 |
| pi3x | depth_x1.5 | drop | 1.0 |
| pi3x | drift_level2 | clean | 1.0 |
| pi3x | drift_level2 | drop | 1.0 |
| pi3x | one_cam_15 | clean | 1.0 |
| pi3x | one_cam_15 | drop | 1.0 |
| pi3x | shared_heading_5 | clean | 1.0 |
| pi3x | shared_heading_5 | drop | 1.0 |

▶ M3 share removed per layer (source: correct-hint run)

▶ M5: swapping the OTHER cameras' states

▶ M2/M4: size of the bad − correct state difference and its cosine with the first layer

### M8: two hints disagree (output scale vs ground truth)

| model | condition | pose_hint_scale | depth_hint_scale | output_scale_vs_gt | ci_lo | ci_hi | shape_err |
|---|---|---|---|---|---|---|---|
| mapanything | clean_pose_depth2 | – | – | 1.025 | 1.010 | 1.041 | 0.031 |
| mapanything | m8_pose0.67_depth0.67 | 0.67 | 0.67 | 0.713 | 0.680 | 0.740 | 0.059 |
| mapanything | m8_pose0.67_depth1.0 | 0.67 | 1.00 | 0.824 | 0.771 | 0.873 | 0.106 |
| mapanything | m8_pose0.67_depth1.5 | 0.67 | 1.50 | 1.058 | 0.971 | 1.141 | 0.191 |
| mapanything | m8_pose1.0_depth0.67 | 1.00 | 0.67 | 0.943 | 0.918 | 0.963 | 0.066 |
| mapanything | m8_pose1.0_depth1.0 | 1.00 | 1.00 | 1.025 | 1.010 | 1.041 | 0.031 |
| mapanything | m8_pose1.0_depth1.5 | 1.00 | 1.50 | 1.157 | 1.126 | 1.190 | 0.077 |
| mapanything | m8_pose1.5_depth0.67 | 1.50 | 0.67 | 1.121 | 1.039 | 1.208 | 0.209 |
| mapanything | m8_pose1.5_depth1.0 | 1.50 | 1.00 | 1.319 | 1.267 | 1.367 | 0.086 |
| mapanything | m8_pose1.5_depth1.5 | 1.50 | 1.50 | 1.450 | 1.426 | 1.474 | 0.034 |
| mapanything | rgb_only | – | – | 1.179 | 1.069 | 1.309 | 0.164 |
| pi3x | clean_pose_depth2 | – | – | 0.887 | 0.744 | 1.017 | 0.013 |
| pi3x | m8_pose0.67_depth0.67 | 0.67 | 0.67 | 0.887 | 0.744 | 1.017 | 0.013 |
| pi3x | m8_pose0.67_depth1.0 | 0.67 | 1.00 | 0.879 | 0.736 | 1.010 | 0.018 |
| pi3x | m8_pose0.67_depth1.5 | 0.67 | 1.50 | 0.885 | 0.741 | 1.016 | 0.085 |
| pi3x | m8_pose1.0_depth0.67 | 1.00 | 0.67 | 0.909 | 0.766 | 1.041 | 0.022 |
| pi3x | m8_pose1.0_depth1.0 | 1.00 | 1.00 | 0.887 | 0.744 | 1.017 | 0.013 |
| pi3x | m8_pose1.0_depth1.5 | 1.00 | 1.50 | 0.879 | 0.736 | 1.009 | 0.018 |
| pi3x | m8_pose1.5_depth0.67 | 1.50 | 0.67 | 0.895 | 0.753 | 1.027 | 0.206 |
| pi3x | m8_pose1.5_depth1.0 | 1.50 | 1.00 | 0.910 | 0.766 | 1.041 | 0.022 |
| pi3x | m8_pose1.5_depth1.5 | 1.50 | 1.50 | 0.887 | 0.744 | 1.017 | 0.013 |
| pi3x | rgb_only | – | – | 0.836 | 0.700 | 0.959 | 0.094 |

## 7. E3 method evaluation (experiments/E3.md, eval splits)

E3 decision (fixed before the run): method claim holds = False; SAFE = True; HELP on agreeing / AI-depth: c6_shared_rot_rot_deg=5, dev_pose_ai_depth, drift_level=2, drift_level=3, shared_heading_bias_rot_deg=10, shared_heading_bias_rot_deg=5; BEST on agreeing / AI-depth: none.

Caution: SAFE passes under the strict rule, but 13 SAFE tests show harm whose 95% CI excludes 0 and whose estimate exceeds the practical threshold, without surviving Holm: real_room gt_pose shape: +0.059 (CI 0.011 to 0.136); real_room gt_pose rotation: +4.032 (CI 0.935 to 7.759); real_room gt_pose real_size: +0.063 (CI 0.012 to 0.144); real_room gt_all shape: +0.070 (CI 0.017 to 0.153); real_room gt_all rotation: +4.073 (CI 0.969 to 7.803); real_room gt_all real_size: +0.073 (CI 0.017 to 0.157); real_room dev_pose shape: +0.055 (CI 0.007 to 0.130); real_room dev_pose rotation: +3.833 (CI 0.767 to 7.515); real_room dev_pose real_size: +0.060 (CI 0.009 to 0.141); real_room dev_all shape: +0.061 (CI 0.009 to 0.142); real_room dev_all rotation: +3.212 (CI

0.411 to 6.918); real_room dev_all real_size: +0.062 (CI 0.009 to 0.144); real_e2e gt_all rotation: +0.984 (CI 0.002 to 2.775). E3 is therefore not shown to be safe on these data.

## SAFE

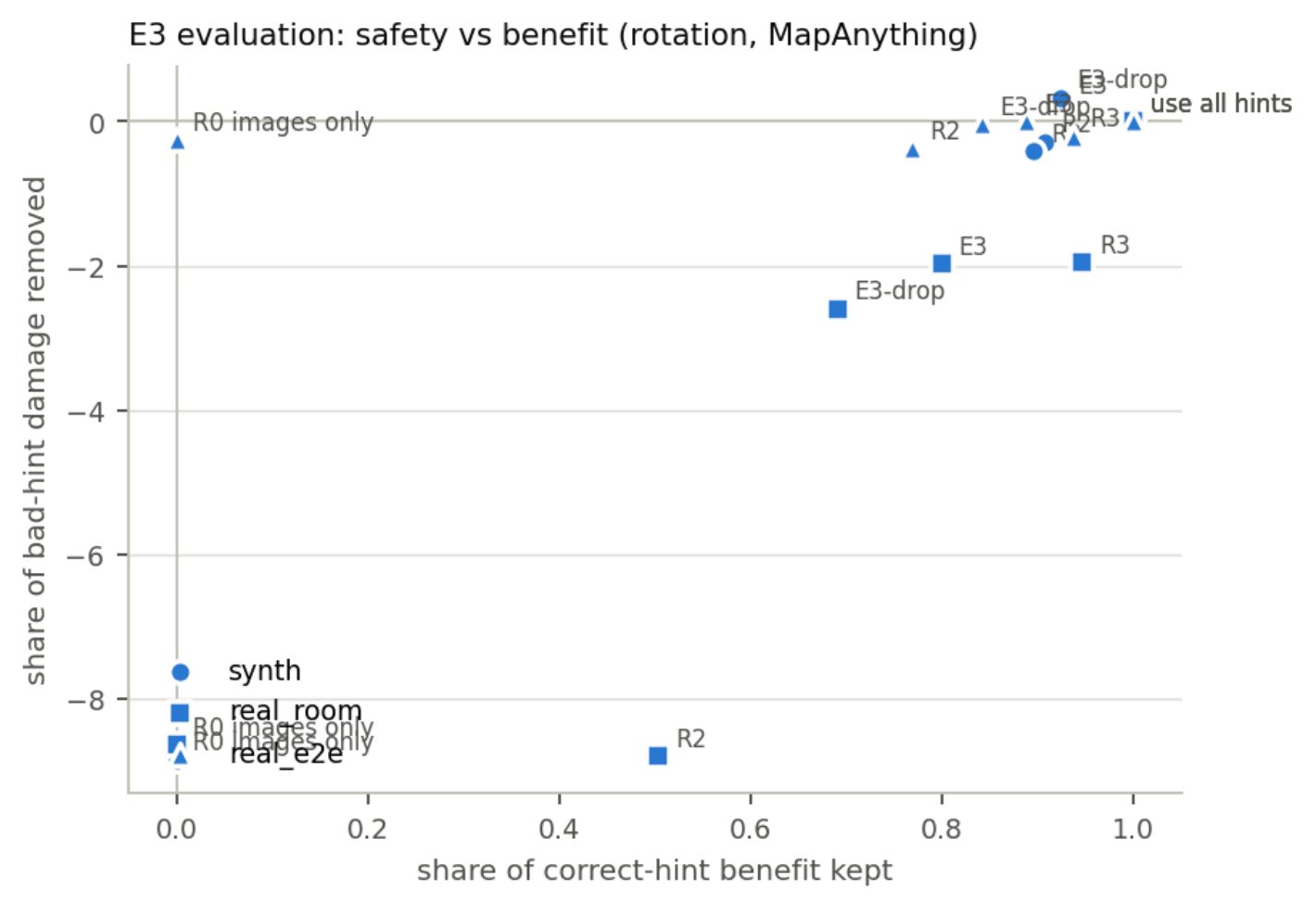
| data | condition | metric | vs | delta | ci_lo | ci_hi | n_scenes | p_holm | harmful |
|---|---|---|---|---|---|---|---|---|---|
| synth | clean_pose | shape | hints as given | 0.001 | 0.000 | 0.002 | 10 | 1.0000 | False |
| synth | clean_pose | rotation | hints as given | 0.110 | 0.000 | 0.221 | 10 | 1.0000 | False |
| synth | clean_pose_depth2 | shape | hints as given | 0.001 | 0.000 | 0.003 | 10 | 1.0000 | False |
| synth | clean_pose_depth2 | rotation | hints as given | 0.116 | 0.000 | 0.238 | 10 | 1.0000 | False |
| real_room | gt_pose | shape | hints as given | 0.059 | 0.011 | 0.136 | 15 | 1.0000 | False |
| real_room | gt_pose | rotation | hints as given | 4.032 | 0.935 | 7.759 | 15 | 1.0000 | False |
| real_room | gt_pose | real_size | hints as given | 0.063 | 0.012 | 0.144 | 15 | 1.0000 | False |
| real_room | gt_all | shape | hints as given | 0.070 | 0.017 | 0.153 | 15 | 1.0000 | False |
| real_room | gt_all | rotation | hints as given | 4.073 | 0.969 | 7.803 | 15 | 1.0000 | False |
| real_room | gt_all | real_size | hints as given | 0.073 | 0.017 | 0.157 | 15 | 1.0000 | False |
| real_room | dev_pose | shape | hints as given | 0.055 | 0.007 | 0.130 | 15 | 1.0000 | False |
| real_room | dev_pose | rotation | hints as given | 3.833 | 0.767 | 7.515 | 15 | 1.0000 | False |
| real_room | dev_pose | real_size | hints as given | 0.060 | 0.009 | 0.141 | 15 | 1.0000 | False |
| real_room | dev_all | shape | hints as given | 0.061 | 0.009 | 0.142 | 15 | 1.0000 | False |
| real_room | dev_all | rotation | hints as given | 3.212 | 0.411 | 6.918 | 15 | 1.0000 | False |
| real_room | dev_all | real_size | hints as given | 0.062 | 0.009 | 0.144 | 15 | 1.0000 | False |
| real_e2e | gt_pose | shape | hints as given | 0.003 | 0.000 | 0.007 | 15 | 1.0000 | False |
| real_e2e | gt_pose | rotation | hints as given | 1.015 | 0.000 | 2.887 | 15 | 1.0000 | False |
| real_e2e | gt_pose | real_size | hints as given | 0.000 | -0.007 | 0.008 | 15 | 1.0000 | False |
| real_e2e | gt_all | shape | hints as given | 0.010 | 0.001 | 0.021 | 15 | 1.0000 | False |
| real_e2e | gt_all | rotation | hints as given | 0.984 | 0.002 | 2.775 | 15 | 1.0000 | False |
| real_e2e | gt_all | real_size | hints as given | 0.018 | 0.001 | 0.045 | 15 | 1.0000 | False |
| real_e2e | dev_pose | shape | hints as given | 0.000 | -0.000 | 0.000 | 15 | 1.0000 | False |
| real_e2e | dev_pose | rotation | hints as given | 0.054 | -0.000 | 0.150 | 15 | 1.0000 | False |
| real_e2e | dev_pose | real_size | hints as given | -0.000 | -0.001 | 0.000 | 15 | 1.0000 | False |
| real_e2e | dev_all | shape | hints as given | 0.004 | -0.000 | 0.011 | 15 | 1.0000 | False |
| real_e2e | dev_all | rotation | hints as given | 0.052 | -0.002 | 0.146 | 15 | 1.0000 | False |
| real_e2e | dev_all | real_size | hints as given | 0.011 | -0.000 | 0.034 | 15 | 1.0000 | False |

## HELP

| data | condition | metric | vs | delta | ci_lo | ci_hi | n_scenes | p_holm | b |
|---|---|---|---|---|---|---|---|---|---|
| synth | outlier_rot_rot_deg=15 | rotation | hints as given | -2.261 | -2.612 | -1.811 | 10 | 0.0002 | T |
| synth | outlier_rot_rot_deg=30 | rotation | hints as given | -5.261 | -5.590 | -4.767 | 10 | 0.0000 | T |
| synth | outlier_rot_rot_deg=90 | rotation | hints as given | -17.552 | -17.640 | -17.459 | 10 | 0.0000 | T |
| synth | outlier_frac_frac=0.2 | rotation | hints as given | -5.057 | -5.207 | -4.882 | 10 | 0.0000 | T |
| synth | outlier_frac_frac=0.4 | rotation | hints as given | -8.955 | -9.357 | -8.468 | 10 | 0.0000 | T |
| synth | shared_heading_bias_rot_deg=5 | rotation | hints as given | -2.308 | -2.900 | -1.773 | 10 | 0.0017 | T |
| synth | shared_heading_bias_rot_deg=10 | rotation | hints as given | -5.665 | -6.477 | -4.889 | 10 | 0.0000 | T |
| synth | c6_shared_rot_rot_deg=5 | rotation | hints as given | -2.431 | -2.964 | -1.958 | 10 | 0.0004 | T |
| synth | drift_level=2 | rotation | hints as given | -2.068 | -2.433 | -1.685 | 10 | 0.0002 | T |
| synth | drift_level=3 | rotation | hints as given | -4.546 | -5.037 | -4.061 | 10 | 0.0000 | T |
| synth | depth_scale_factor=0.67 | shape | hints as given | -0.032 | -0.043 | -0.021 | 10 | 0.0222 | T |
| synth | depth_scale_factor=1.5 | shape | hints as given | -0.042 | -0.059 | -0.027 | 10 | 0.0494 | T |
| synth | depth_one_scale_factor=1.5 | shape | hints as given | -0.003 | -0.008 | 0.002 | 10 | 1.0000 | F |
| real_room | ai_depth | shape | hints as given | -0.044 | -0.101 | 0.015 | 15 | 1.0000 | F |
| real_room | dev_pose_ai_depth | shape | hints as given | -0.066 | -0.110 | -0.000 | 15 | 1.0000 | F |
| real_e2e | ai_depth | shape | hints as given | -0.074 | -0.115 | -0.041 | 15 | 0.1002 | F |
| real_e2e | dev_pose_ai_depth | shape | hints as given | -0.061 | -0.077 | -0.046 | 15 | 0.0001 | T |

**BEST**

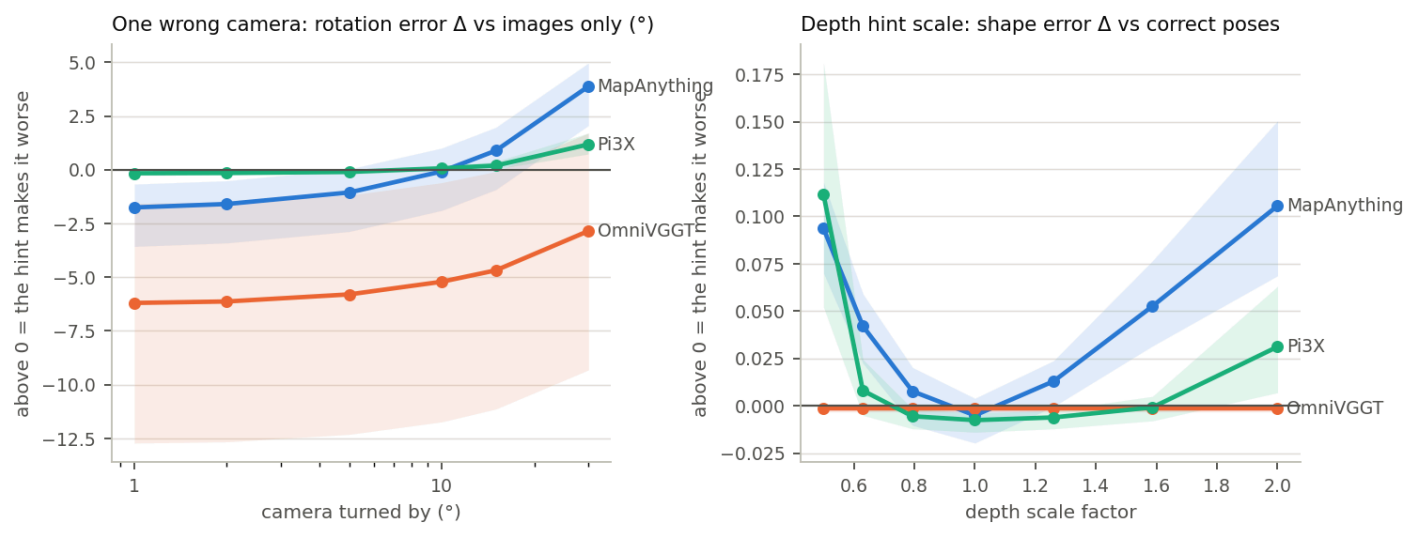
| data | condition | metric | vs | delta | ci_lo | ci_hi | n_scenes | p_holm | bet |
|---|---|---|---|---|---|---|---|---|---|
| synth | outlier_rot_rot_deg=15 | rotation | filter_rgbcheck | 0.307 | 0.023 | 0.668 | 10 | 1.0000 | Fals |
| synth | outlier_rot_rot_deg=30 | rotation | filter_rgbcheck | 0.228 | -0.058 | 0.683 | 10 | 1.0000 | Fals |
| synth | outlier_rot_rot_deg=90 | rotation | filter_rgbcheck | -0.033 | -0.151 | 0.085 | 10 | 1.0000 | Fals |
| synth | outlier_frac_frac=0.2 | rotation | filter_rgbcheck | -0.050 | -0.174 | 0.081 | 10 | 1.0000 | Fals |
| synth | outlier_frac_frac=0.4 | rotation | filter_rgbcheck | 0.604 | 0.255 | 1.012 | 10 | 0.7147 | Fals |
| synth | shared_heading_bias_rot_deg=5 | rotation | rgb_only | -1.394 | -3.057 | -0.488 | 10 | 1.0000 | Fals |
| synth | shared_heading_bias_rot_deg=10 | rotation | rgb_only | -1.629 | -3.478 | -0.582 | 10 | 1.0000 | Fals |
| synth | c6_shared_rot_rot_deg=5 | rotation | rgb_only | -1.272 | -2.998 | -0.240 | 10 | 1.0000 | Fals |
| synth | drift_level=2 | rotation | rgb_only | -0.865 | -2.504 | 0.078 | 10 | 1.0000 | Fals |
| synth | drift_level=3 | rotation | rgb_only | 0.037 | -1.738 | 1.193 | 10 | 1.0000 | Fals |
| synth | depth_scale_factor=0.67 | shape | drop_lowconf | -0.031 | -0.042 | -0.021 | 10 | 0.0168 | True |
| synth | depth_scale_factor=1.5 | shape | drop_lowconf | -0.041 | -0.059 | -0.024 | 10 | 0.0826 | Fals |
| synth | depth_one_scale_factor=1.5 | shape | drop_lowconf | 0.000 | -0.004 | 0.005 | 10 | 1.0000 | Fals |
| real_room | ai_depth | shape | rgb_only | 0.009 | -0.041 | 0.049 | 15 | 1.0000 | Fals |
| real_room | dev_pose_ai_depth | shape | drop_lowconf | -0.059 | -0.102 | 0.006 | 15 | 1.0000 | Fals |
| real_e2e | ai_depth | shape | rgb_only | 0.026 | 0.018 | 0.034 | 15 | 0.0011 | Fals |
| real_e2e | dev_pose_ai_depth | shape | rgb_only | 0.024 | 0.015 | 0.033 | 15 | 0.0106 | Fals |



| data | metric | remedy | benefit_kept | damage_removed | n_bad_conditions |
|---|---|---|---|---|---|
| synth | rre_deg | none | 1.000 | 0.000 | 13 |
| synth | rre_deg | witness | 0.925 | 0.266 | 13 |
| synth | rre_deg | witness_drop | 0.923 | 0.331 | 13 |
| synth | rre_deg | filter_rgbcheck | 0.907 | -0.280 | 13 |
| synth | rre_deg | drop_lowconf | 0.896 | -0.412 | 13 |
| synth | rre_deg | drop_oracle | – | 0.971 | 5 |
| synth | rre_deg | rgb_only | 0.000 | -8.840 | 13 |
| synth | pts_relerr_scale | none | 1.000 | 0.000 | 13 |
| synth | pts_relerr_scale | witness | 0.659 | 0.773 | 13 |
| synth | pts_relerr_scale | witness_drop | 1.214 | 0.111 | 13 |
| synth | pts_relerr_scale | filter_rgbcheck | 0.823 | -1.476 | 13 |
| synth | pts_relerr_scale | drop_lowconf | 0.938 | 0.179 | 13 |
| synth | pts_relerr_scale | drop_oracle | – | 0.939 | 5 |
| synth | pts_relerr_scale | rgb_only | 0.000 | -2.074 | 13 |
| real_room | rre_deg | none | 1.000 | 0.000 | 2 |
| real_room | rre_deg | witness | 0.800 | -1.966 | 2 |
| real_room | rre_deg | witness_drop | 0.691 | -2.591 | 2 |
| real_room | rre_deg | filter_rgbcheck | 0.503 | -8.787 | 1 |
| real_room | rre_deg | drop_lowconf | 0.946 | -1.954 | 1 |
| real_room | rre_deg | drop_oracle | – | – | 0 |
| real_room | rre_deg | rgb_only | 0.000 | -8.627 | 2 |
| real_room | pts_relerr_scale | none | 1.000 | 0.000 | 2 |
| real_room | pts_relerr_scale | witness | 0.718 | 0.302 | 2 |
| real_room | pts_relerr_scale | witness_drop | 0.709 | 0.365 | 2 |
| real_room | pts_relerr_scale | filter_rgbcheck | 0.434 | -0.332 | 1 |
| real_room | pts_relerr_scale | drop_lowconf | 0.929 | 0.044 | 1 |
| real_room | pts_relerr_scale | drop_oracle | – | – | 0 |
| real_room | pts_relerr_scale | rgb_only | 0.000 | 0.073 | 2 |
| real_e2e | rre_deg | none | 1.000 | 0.000 | 2 |
| real_e2e | rre_deg | witness | 0.888 | 0.010 | 2 |
| real_e2e | rre_deg | witness_drop | 0.842 | -0.032 | 2 |
| real_e2e | rre_deg | filter_rgbcheck | 0.769 | -0.369 | 1 |
| real_e2e | rre_deg | drop_lowconf | 0.937 | -0.207 | 1 |
| real_e2e | rre_deg | drop_oracle | – | – | 0 |
| real_e2e | rre_deg | rgb_only | 0.000 | -0.255 | 2 |
| real_e2e | pts_relerr_scale | none | 1.000 | 0.000 | 2 |
| real_e2e | pts_relerr_scale | witness | 0.959 | 0.647 | 2 |
| real_e2e | pts_relerr_scale | witness_drop | 0.864 | 0.774 | 2 |

| data | metric | remedy | benefit_kept | damage_removed | n_bad_conditions |
|---|---|---|---|---|---|
| real_e2e | pts_relerr_scale | filter_rgbcheck | 0.971 | -0.036 | 1 |
| real_e2e | pts_relerr_scale | drop_lowconf | 0.942 | 0.028 | 1 |
| real_e2e | pts_relerr_scale | drop_oracle | – | – | 0 |
| real_e2e | pts_relerr_scale | rgb_only | 0.000 | 0.888 | 2 |

## 5. Angles (DESCRIPTIVE)




Smallest error size whose 95% CI lies above 0 (the hint is then worse than going without it):

| model | family | condition | size | delta_vs_floor | ci_lo | ci_hi |
|---|---|---|---|---|---|---|
| mapanything | c1_depth | c1_depth_factor=0.5 | 0.50 | 0.094 | 0.070 | 0.118 |
| mapanything | c1_rot | c1_rot_rot_deg=30 | 30.00 | 3.875 | 2.057 | 4.959 |
| pi3x | c1_depth | c1_depth_factor=0.5 | 0.50 | 0.111 | 0.052 | 0.181 |
| pi3x | c1_rot | c1_rot_rot_deg=15 | 15.00 | 0.194 | 0.035 | 0.379 |
| pi3x | c1_trans | c1_trans_trans_frac=3 | 3.00 | 0.284 | 0.137 | 0.420 |

▶ All break-even rows

## 6. Geometry referee (L2, planned baseline R5; dev/cal clips)

Calibration (rule fixed in the script): view part **NOT CALIBRATED**, set part **NOT CALIBRATED**; 17% of views scored. Details: `results/e2x_geometry_referee_deg/README.md`.

ETH3D dev, one wrong camera: the referee measures the error itself.

| condition | bad_view_median_deg | good_view_median_deg |
|---|---|---|
| outlier_rot_rot_deg=15 | 14.97 | 1.01 |
| outlier_rot_rot_deg=30 | 29.86 | 0.95 |
| outlier_rot_rot_deg=5 | 5.17 | 0.82 |
| outlier_rot_rot_deg=90 | 89.96 | 1.01 |

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## 8. Files

- `results/e2x_agg/angle_break_even.csv`
- `results/e2x_agg/angle_safety_preview.csv`
- `results/e2x_agg/xa_behaviour_ma_vs_pi3x.csv`
- `results/e2x_agg/xa_behaviour_pi3x.csv`
- `results/e2x_agg/xa_family_ma_vs_pi3x.csv`
- `results/e2x_agg/xa_harm_pi3x.csv`
- `results/e2x_agg/xb_e2e_tests_pi3x.csv`
- `results/e2x_agg/xb_room_real_errors.csv`
- `results/e2x_agg/xb_room_remedies_mapanything.csv`
- `results/e2x_agg/xb_room_remedies_omnivggt.csv`
- `results/e2x_agg/xb_room_remedies_pi3x.csv`
- `results/e2x_agg/xb_room_tests_mapanything.csv`
- `results/e2x_agg/xb_room_tests_omnivggt.csv`
- `results/e2x_agg/xb_room_tests_pi3x.csv`
- `results/e2x_e2e_px_dev/per_run.csv`
- `results/e2x_e2e_px_eval/per_run.csv`
- `results/e2x_e3cal_agg/mapanything/chosen.json`
- `results/e2x_e3cal_agg/mapanything/detection_rates.csv`
- `results/e2x_e3cal_agg/mapanything/false_action_rates.csv`
- `results/e2x_e3cal_agg/mapanything_secondary/chosen.json`
- `results/e2x_e3cal_agg/mapanything_secondary/false_action_rates.csv`
- `results/e2x_e3cal_agg/pi3x/chosen.json`
- `results/e2x_e3cal_agg/pi3x/detection_rates.csv`
- `results/e2x_e3cal_agg/pi3x/false_action_rates.csv`
- `results/e2x_e3cal_agg/pi3x_secondary/chosen.json`
- `results/e2x_e3cal_agg/pi3x_secondary/false_action_rates.csv`
- `results/e2x_e3cal_real_ma_cal/per_run.csv`
- `results/e2x_e3cal_real_ma_roomdev/per_run.csv`
- `results/e2x_e3cal_real_px_cal/per_run.csv`
- `results/e2x_e3cal_synth_ma_cal/per_run.csv`
- `results/e2x_e3cal_synth_ma_eth3d/per_run.csv`
- `results/e2x_e3cal_synth_ma_tum/per_run.csv`
- `results/e2x_e3cal_synth_ov_eth3d/per_run.csv`
- `results/e2x_e3cal_synth_px_cal/per_run.csv`
- `results/e2x_e3cal_synth_px_eth3d/per_run.csv`
- `results/e2x_e3cal_synth_px_tum/per_run.csv`

- results/e2x_geometry_referee_deg/chosen.json
- results/e2x_geometry_referee_deg/eth3d_dev_descriptive.csv
- results/e2x_pi3x_checks/per_run.csv
- results/e2x_pi3x_eth3d_dev/per_run.csv
- results/e2x_pi3x_eth3d_eval/per_run.csv
- results/e2x_room_ma_dev/per_run.csv
- results/e2x_room_ma_eval/per_run.csv
- results/e2x_room_ov_dev/per_run.csv
- results/e2x_room_ov_eval/per_run.csv
- results/e2x_room_px_dev/per_run.csv
- results/e2x_room_px_eval/per_run.csv
- results/e2x_xm_agg/m1_curve.csv
- results/e2x_xm_agg/m1_dial.csv
- results/e2x_xm_agg/m2_m4_state.csv
- results/e2x_xm_agg/m3_patch.csv
- results/e2x_xm_agg/m3_point_of_no_return.csv
- results/e2x_xm_agg/m5_spread.csv
- results/e2x_xm_agg/m8_scale.csv
- results/e3_eval_agg/e3_safety_benefit.csv
- results/e3_eval_agg/e3_tests.csv
- results/e3_eval_agg/e3_verdict.json
- results/vm_logs_e2x/machine.txt
- results/vm_logs_e2x/mono_verify.txt
- results/vm_logs_e2x/session.txt